



OPEN Development of a stream DTM generation methodology using UAV-based SfM and LiDAR point cloud

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Filtering ground points and generating accurate Digital Terrain Models (DTMs) are crucial for hydrological and ecological modeling in riverine environments. This paper proposes a methodology for DTM generation in stream areas by integrating point clouds from a UAV-based structure from motion (SfM) and light detection and ranging (LiDAR) while applying ground filtering algorithms. The study area, the Bokha Stream in Icheon City, South Korea, was surveyed using Zenmuse L1 (LiDAR) and Phantom 4 multispectral (SfM). Water and non-water areas were classified using the normalized difference water index (NDWI), and three ground filters, cloth simulation filter (CSF), progressive triangular irregular network (PTIN), and simple morphological filter (SMRF), were applied to remove vegetation and error points. The methodology was validated using 11 cross-sections, with accuracy assessed via mean absolute error (MAE) and root mean square error (RMSE). Among the filters, SMRF demonstrated superior performance, achieving an MAE of 0.160 m and an RMSE of 0.214 m. It effectively removed vegetation and preserved riparian terrain but tended to underestimate elevations in water areas. The results highlight the potential of the proposed method to enhance the accuracy of DTMs, offering benefits for applications in mesoscale hydrological and ecological modeling in small streams.

Keywords Structure from motion (SfM), LiDAR (Light detection and ranging), Digital terrain model (DTM), Ground filter, Stream

Acquiring stream topographic information is crucial for accurate hydrological and ecological modeling¹. Some researchers have adopted a cross-sectional surveying method for river modeling, which involves manually measuring each point of the river cross-section using electronic distance measurement (EDM)². The direct survey method allows for precise mapping of river topography, however, it has significant limitations: it is time consuming, costly, poses safety risks, and cannot provide continuous topographic information³. The traditional sonar based methods such as side-scan sonar and multibeam swath systems have been widely used for river and bathymetry area, but they are ineffective in shallow stream which depth is lower than 0.3–0.5 m⁴, and are unsuitable for generating a comprehensive DTM that includes water, riparian vegetation, and the surrounding ground. As an alternative, unmanned aerial vehicles (UAV), which can generate high-resolution digital terrain models (DTMs) in small stream environments, have been widely adopted in recent studies⁵.

To generate a precise DTM by UAV, it is essential to acquire point clouds and extract ground points. There are two main methods for obtaining point clouds using UAV: the light detection and ranging (LiDAR) method and the structure from motion (SfM) method. LiDAR generates point clouds by emitting laser pulses and measuring the time required for the pulses to return after hitting an object⁶. The LiDAR method can partially penetrate the vegetation canopy and record multiple laser pulse returns, making it effective for filtering vegetation and ground points^{4,7}. However, most LiDAR systems use near-infrared wavelengths, typically around 905–1550 nm, which reflect off the surface of water, making it impossible to provide point clouds of the channel bed⁸. Some studies have applied green LiDAR, which uses a green wavelength of 532 nm, to detect the bottom of water bodies⁸, but

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its effectiveness can be significantly affected by the environmental conditions of water and is too expensive to be used widely⁹.

Structure from motion (SfM), also known as the photogrammetry method, operates by capturing multiple overlapping images of an object or scene from different angles and using photogrammetric algorithms to reconstruct a three-dimensional model from these two-dimensional images¹⁰. The SfM method is mainly effective in reconstructing visible surfaces, but has difficulties in acquiring ground points beneath dense canopies⁷. However, if the riverbed remains sufficiently visible, the SfM method can also be used to generate a point cloud of the riverbed, unlike the LiDAR method^{11,12}.

Because the raw point cloud data include not only ground objects but also various non-ground objects such as vegetation, buildings, and poles, it is essential to extract ground points for accurate DTM generation. Filtering ground points remains a challenge because of the irregular shapes and sizes of non-ground objects¹³. Various filtering methods have been investigated to address these challenges. For instance, Axelsson¹⁴ proposed the progressive triangular irregular network (PTIN), which filters the ground points based on TIN interpolation. Pingel et al.¹⁵ introduced a simple morphological filter (SMRF) that uses morphological operations to filter ground points. Zhang et al.¹⁶ proposed a cloth simulation filter (CSF) that employs a cloth simulation algorithm for ground filtering.

Since the introduction of PointNet, several deep learning models such as PointNet++, DGCNN and RandLA-Net have been developed^{17–20}. Ciou et al.²¹ developed an DGCNN based ensemble learning-based method that integrates deep learning classifiers specialized for urban and mountainous terrains to enhance ground point classification accuracy and improve DEM generation from airborne LiDAR point clouds²² developed the Multi-Size Voxel Cube (MSVC) algorithm, a novel deep learning-based terrain filtering method that progressively reduces voxel size to accurately extract ground points from dense and rugged point clouds.

Filtering methods have higher interpretability and are more robust across various datasets^{23,24}. However, the accuracy of filtering methods can vary significantly depending on the parameters used, and they still face challenges in complex terrains such as riverine environments¹⁶. Deep learning based methods generally offer higher accuracy compared to traditional filtering methods, as they can learn complex patterns and relationships in the data¹⁷, however, they have lower reproducibility and the necessity of extensive labeled data for training²⁵.

Furthermore, gathering topographic information in water areas presents additional challenges for constructing stream DTMs. To address this, combining SfM and LiDAR point clouds for water and non-water areas separately can be effective; however, it requires accurate classification of water points. Several indices for water classification have been widely studied owing to the wavelength-dependent reflection properties of water surfaces. McFeeters²⁶ introduced the normalized difference water index (NDWI), using near-infrared (NIR) and green bands, to delineate water areas in the sandhill region. Xu²⁷ introduced the modified NDWI (MNDWI), using short wavelength infrared (SWIR) and green bands, and calculated different thresholds of NDWI and MNDWI indices for lakes, streams, and seas.

Several studies have employed UAV-based approaches to generate digital terrain models (DTMs) in stream environments. Tan et al.²⁸ applied filters such as CSF and PTIN to SfM-derived point clouds for riverbank monitoring, however, their analysis was limited to terrestrial areas and did not estimate water depth. Szabó et al.²⁹ generated a DTM by applying the CSF algorithm to LiDAR point clouds in fluvial areas, but the study lacked comparative evaluation with other filtering algorithms. Szombara et al.³⁰ combined LiDAR data and orthophotomaps to produce DTMs in stream environments, however, dense riparian vegetation and the inability of LiDAR to penetrate the water surface constrained the completeness of the derived DTMs. Recent research has explored the integration of SfM and LiDAR for more accurate DTM generation. Conesa-García et al.³¹ produced DTMs for gravel-bed streams by combining SfM photogrammetry and terrestrial laser scanning, however, they did not compare the performance of different filtering algorithms or evaluate accuracy in submerged areas. Bi et al.⁶ used UAV-LiDAR and SfM data to generate high-resolution DTMs in mountainous regions for geomorphic analysis, but their study excluded stream channels. Rusnák et al.¹² generated DTMs for stream environments using a combined SfM–LiDAR approach and applied a water refraction correction method, however, they manually defined water surface boundaries and did not consider the effects of ground-filtering parameters on DTM accuracy.

The objective of this paper is to propose a robust methodology for generating high-precision DTMs in stream areas by integrating UAV-based SfM and LiDAR point clouds. To reflect the different performance of SfM and LiDAR, using SfM for acquiring waterbed points and using LiDAR for penetrating vegetation in riparian zones, we implemented an automated water area classification using the NDWI. Furthermore, this study evaluated the performance of various ground-filtering algorithms by testing various parameter combinations. Through this comprehensive optimization, we aim to identify the most effective strategy for accurate terrain modeling in complex riparian environments.

Materials and methods

Figure 1 shows the overall process of generating the DTM using the LiDAR and SfM methods, followed by an accuracy evaluation using EDM survey data. Initially, point clouds were generated using LiDAR and photogrammetry. The NDWI of each pixel was also calculated using a photogrammetry survey. Ground-truth cross-sections were acquired using a cross-section survey, using total station and GPS module, including boundary points of ground, vegetation, and water areas. Point clouds were preprocessed by down-sampling and water and non-water extraction using NDWI. Both the SfM and LiDAR point clouds were filtered using three ground filters and integrated to generate the final DTM with TIN interpolation. Digital surface model (DSM) was also generated by selecting the highest elevation at 5 cm/pixel. Finally, the mean absolute error (MAE) and root mean square error (RMSE) were used to evaluate all DTMs.

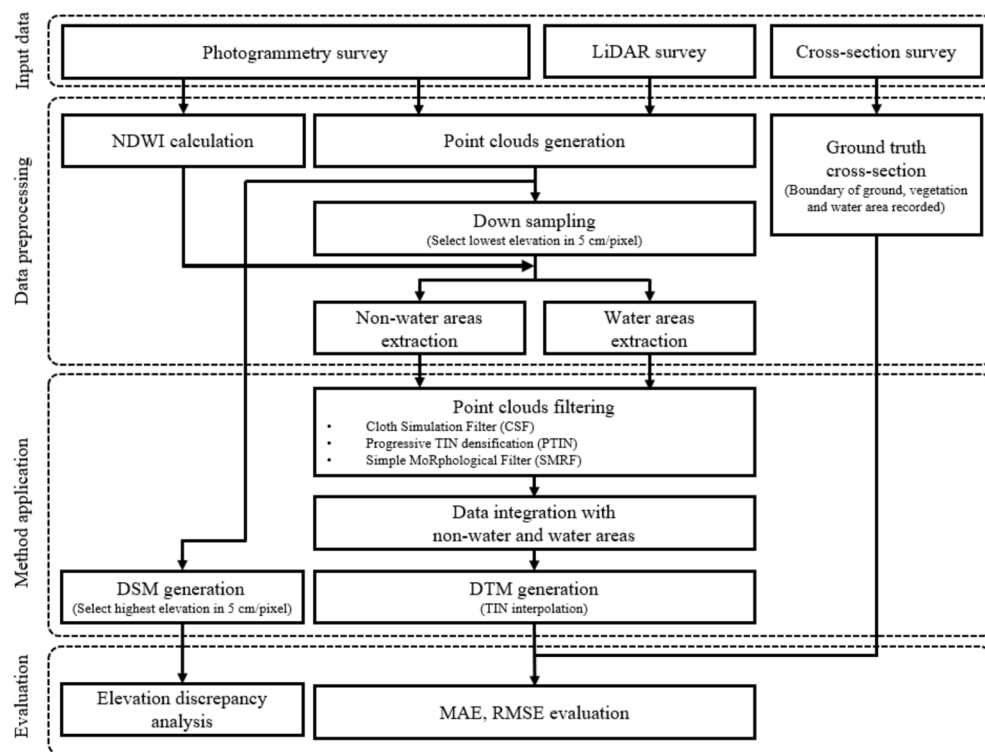


Fig. 1. Process of generating DTMs and accuracy evaluation using LiDAR and SfM point clouds.

Study area and image acquisition

The study area was a reach of the Bokha stream in Icheon, South Korea, which spans 2.8 km in length, approximately 250 m in width, and an area of 3.95 km² (Figure 2). The Bokha stream spans a total length of 39.84 km, and has a basin area of 309.5 km²³². Because of the high flow fluctuation index, which is 875 in the Bokha stream compared to 30 in the Nile River, it has a complex topography, including riverbeds, sandbanks, floodplains, and embankments³². The stream also supports a diverse ecology, with tall trees such as *Robina pseudo-acacia* L. et al., planted along the top of the embankments, and low shrubs such as *Phragmites japonica* Steud. inhabiting inland of the stream, as seen in the site image in Figure 2.

Drone and EDM surveys were conducted from February 26, 2024, to February 28, 2024, when the water depth and vegetation vitality were low. SfM data were acquired using a Phantom 4 multispectral drone (DJI, China, Figure 3 (a)), which provides red, green, blue, red-edge, and NIR bands. The images were captured with a resolution of 5 cm/pixel, flight speed of 5 m/s, flight altitude of 95 m, longitudinal overlap of 75%, and lateral overlap of 60%. 63 GCPs, of 1 m × 1 m were placed every 100–150 m for georeferencing (Figure 2). The longitude, latitude, and altitude of each GCP were measured using a GPS module (GeoXR, Trimble, US, Figure 3 (c)) with Real-Time Kinematic (RTK) surveying, using Virtual Reference Station (VRS). Pix4Dmapper (ver. 4.8.4) was used for image registration. The generated point cloud has a density of 288 points/m².

LiDAR data were acquired using a Matrice 300 drone (DJI, China) with a Zenmuse L1 module (Figure 3 (b)), which can receive three reflection signals with a single transmission. The flight was conducted at an altitude of 110 m, a speed of 9.5 m/s, a longitudinal overlap of 70%, and a lateral overlap of 60%. DJI Terra (ver. 3.7.6) was used for postprocessing. The point-cloud density was set to 181 points/m². The average elevation difference between SfM and LiDAR for each GCP was 0.004 m, with a standard deviation of 0.103 m.

Eleven cross-sections were measured using EDM (ES-52, Topcon, Figure 3 (d)) and GPS module, which are represented by red lines and numbered with white circles in Figure 2. Due to the impracticality of bringing expensive GPS into the river, only the endpoints of the cross-section were surveyed using GPS to obtain absolute coordinates, while the cross-sectional profiles were measured using an EDM. The surveyor recorded all points perceived as inflection points and marked the boundaries between vegetation and ground, as well as between ground and water surfaces. During the measurement, we operated in full compliance with all relevant national and local regulations.

Point clouds down sampling and DSM generation

Applying a ground filtering algorithm to billions of point clouds collected from large areas is computationally intensive and time-consuming without adequate preprocessing. Moreover, because LiDAR detects multiple returns from a single transmitted pulse, point cloud density can vary across different objects, leading to inefficiencies in generating DTMs³³. To enhance the accuracy, we selected the last return from the LiDAR data. Additionally, to reduce the processing time and ensure uniform point cloud density, we performed downscaling on both SfM and LiDAR data by extracting the lowest elevation point within 5 cm pixels. Preprocessing resulted

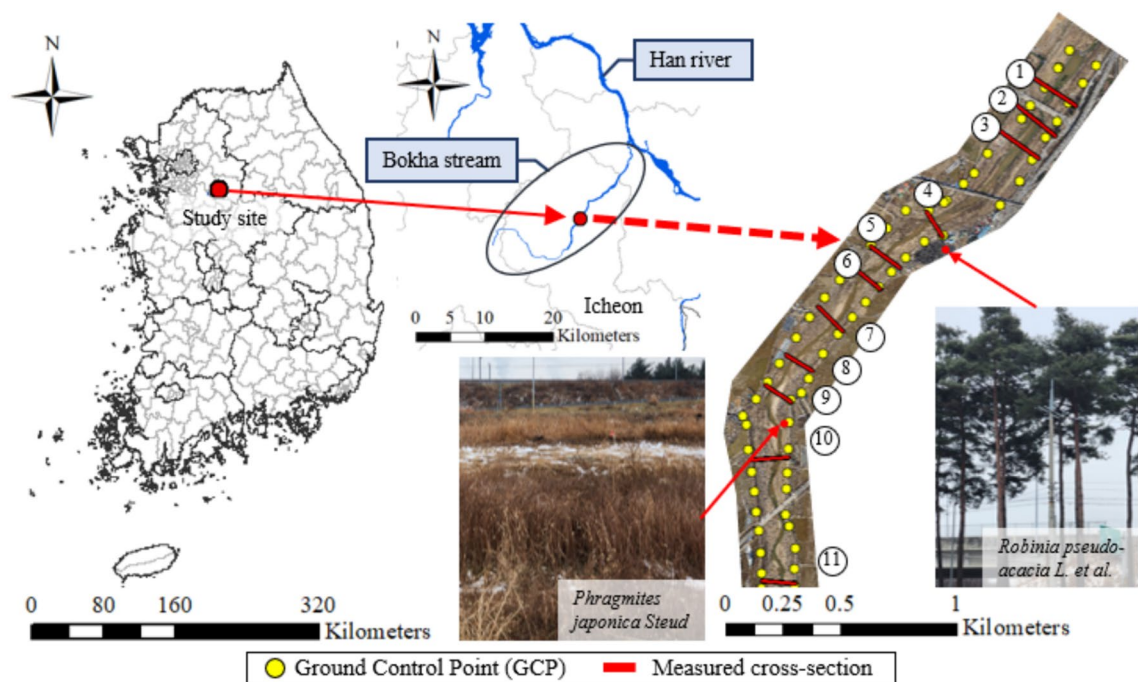


Fig. 2. Location of cross-sections in Bokha stream, Korea. The red line indicates the measured cross-section and yellow dots indicates the GCPs. The numbers in the white circle represents the cross-section number which surveyed by EDM. The aerial map in the left side is drone captured orthomosaics using Phantom 4 multispectral. The photos are captured by authors during cross-section measurement. This map was edited using ArcMap 10.1 (ESRI, <https://desktop.arcgis.com/en/arcmap/latest/get-started/main/get-started-with-arcmap.htm>).



Fig. 3. On-site pictures of drones and devices used in this study. (a) indicates Phantom 4 multispectral drone, (b) indicates Zenmuse L1 drone, (c) indicates GCP measurements using Trimble GeoXR RTK GPS module, and (d) indicates cross-section measurements using Topcon ES-52 EDM module.

in a 25% reduction in the number of points for SfM point clouds and a 43% reduction for LiDAR point clouds (Table 1).

Simultaneously, we also generated a Digital Surface model (DSM) by extracting the highest elevation point within 5 cm pixels and conducting TIN interpolation. The DSM was primarily used for visual comparison with the generated DTM and to assess the performance of ground filtering algorithms by highlighting differences in surface representation. As LiDAR cannot represent the water area, we used SfM point clouds to generate the DSM. The mean elevation difference between the down sampled SfM and DSM was 0.224 m.

Water areas filtering

LiDAR provides high accuracy in vegetation areas, whereas SfM can overcome the shortcomings of LiDAR in water areas^{7,34}. To effectively combine LiDAR and SfM, it is essential to distinguish between water and non-water areas. In this study, we used NDWI, which offers satisfactory performance and is compatible with the specifications of the Phantom 4 Multispectral. NDWI was calculated using equation (1).

NDWI = (NIR - GREEN) / (NIR + GREEN) (1)

We used the index calculator in Pix4Dmapper to calculate the NDWI for every pixel in the drone-captured images. The threshold applied was 0.19, as recommended by Xu²⁷ for shallow, turbid riverine area. After classifying the pixels into water and non-water regions, LiDAR point clouds from non-water areas and SfM point clouds from water areas were selected to construct riverine DTMs.

Ground areas filtering algorithm

LiDAR point clouds capture various objects, including the ground, vegetation, electric poles, and vehicles, while also containing noise introduced by high humidity in the air. In addition, SfM point clouds are particularly susceptible to inaccuracies in water areas caused by turbulence, suspended particles, and reflectance of the water surface^{35,26}. To eliminate error points and accurately classify ground points, we employed three ground filtering algorithms: CSF, PTIN, and SMRF, whose performance has been widely validated in several studies^{7,23,28}.

CSF

The Cloth Simulation Filter (CSF) models a virtual cloth that falls onto the inverted point cloud under gravity, and its equilibrium configuration represents the ground surface. The motion of each cloth particle follows the mass-spring equation, as described as equation (2).

m_i * d^2 x_i / dt^2 = F_i^ext + F_i^int + F_i^col (2)

Where m_i is the mass of the i-th particle, x_i is its position vector, F_i^ext is the external force, F_i^int is the internal elastic forces from neighboring particles, and F_i^col is the collision force exerted by the point cloud surface. The CSF is widely recognized for its computational simplicity, terrain adaptability, and robustness against noise and outliers^{37,16}.

The CSF uses five parameters: resolution, rigidity, threshold, maximum iteration, and slope processing. The resolution determines the mesh spacing between particles. The rigidity controls stiffness of the springs that generate the F_i^int and the threshold defines the maximum vertical distance between a point in the original point cloud and the equilibrium cloth surface used for classification^{16,37}. The maximum iteration specifies the number of iterations for the simulation, and slope processing is a Boolean flag (0 or 1) determining whether local slope adjustments are applied to F_i^col¹⁶. Higher resolution along with appropriately set rigidity and threshold values can improve the accuracy of ground point classification but also increase computational demands¹⁶.

In this study, the CSF was applied using the CSF library (ver. 1.1.1) using Python (ver. 3.8.18). Rigidity was set to 1, a value commonly used for steep slopes, and max iteration to the recommended value of 500³⁸. Slope processing was set to false to minimize the risk of misclassifying vegetation-induced undulations as terrain features. The experimental set of parameters was the same as that in Table 2.

PTIN

The progressively triangulated irregular network (PTIN), which improved the adaptive triangulated irregular network (ATIN) of Axelsson¹⁴. The ATIN begins by constructing an initial ground surface using the lowest elevation points in the LiDAR dataset. Each remaining point is then tested against the current triangulated surface to determine whether it should be classified as ground. For each candidate, two geometric constraints are applied as equation (3).

Measured method	The number of point clouds	
	Before preprocessing	After preprocessing
SfM	408,981,071	305,993,229
LiDAR	475,893,957	271,992,362

Table 1. The decrement of the number of SfM and LiDAR point clouds after the preprocessing.

Algorithm	Parameter	Value experimented
CSF	Threshold (m)	0.01, 0.02, 0.03, 0.04, 0.05, 0.06
	Rigidness (m)	1
	Resolution (m)	0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7
	Slope Processing	FALSE
	Max iteration	500 times
PTIN	Step (m)	3, 4, 5, 6, 7
	Spike (m)	0.2, 0.4, 0.6, 0.8, 1.0
	Offset (m)	0.01, 0.02, 0.03, 0.04, 0.05
	Bulge (m)	1
	Stddev	FALSE
	Sub	extra fine
SMRF	Cell (m)	1.0
	Window (m)	1, 2, 3, 4, 5
	Slope	0.02, 0.04, 0.06, 0.08, 0.10
	Scalar	0.1, 0.2, 0.3, 0.4, 0.5, 0.6
	Threshold (m)	0.01, 0.02, 0.03, 0.04, 0.05, 0.075, 0.10

Table 2. The set of parameters experimented for each algorithm. The range of the value were referred by Asghar³⁹, Cai et al.³⁷, Pingel et al.¹⁵.

$$|z_k - z_T(x_k, y_k)| \leq h_{\max}(d_k), \quad \theta_k \leq \theta_{\max}(d_k) \quad (3)$$

Where $z_T(x_k, y_k)$ is the interpolated elevation of the TIN at the point's horizontal location, d_k is the horizontal distance from the nearest TIN node, $h_{\max}(d_k)$ is the adaptive height threshold, and $\theta_{\max}(d_k)$ is the adaptive maximum slope angle. If both conditions are satisfied, remaining point is added to the ground set, and the TIN is updated accordingly. By iteratively applying these rules, the algorithm adaptively expands the ground surface, allowing larger thresholds in rougher terrain and smaller ones in flatter areas.

The PTIN, provided by Lasground software, an open-source program made by Rapidlasso Co, has similar procedure with ATIN algorithm controlled by controlled by six parameters: step, sub, bulge, spike thresholds, ground offset and standard deviation. In initial ground surface construction procedure, step parameter defines the size of grid and sub parameter determines the level of grid subdivision, which refines the density of initial ground candidates depending on terrain steepness. In iterative TIN procedure, bulge parameter determines the $h_{\max}(d_k)$ and points exceeding the spike threshold are excluded as outlier. After the TIN converges, points are classified as ground if they lie within the offset distance and standard deviation from the final TIN surface³⁹.

Following recommendations from previous studies, we set Bulge and Stddev to their default values, used extra-fine for Sub to reflect complex terrain variations, and experimented with Step, Spike, and Offset values, as shown in Table 2³⁹.

SMRF

The Simple Morphological Filter (SMRF) utilizes morphological opening including erosion and delation, which is described as equation (4).

$$O_B(f) = (f \ominus B) \oplus B \quad (4)$$

Where $f(x, y)$ is the input surface and B is the structuring element. The erosion $\ominus$ and dilation $\oplus$ are respectively defined as equation (5) and (6).

$$(f \ominus B)(x, y) = \min_{(s, t) \in B} f(x + s, y + t) \quad (5)$$

$$(f \oplus B)(x, y) = \max_{(s, t) \in B} f(x - s, y - t) \quad (6)$$

$O_B(f)$, calculated by morphological opening, represents the ground surface, and filters ground points by elevation difference between the ground surface¹⁵. The SMRF is known for its simplicity and computational efficiency, making it suitable for both urban and rural landscapes. However, SMRF is less effective in highly uneven terrains and may require careful parameter tuning to avoid oversmoothing^{7,15}.

The SMRF parameters include the Cell, Slope, Window, Threshold, and Scalar. The cell defines the grid spacing for processing, where each grid cell stores the minimum elevation. The slope parameter contains the allowable elevation difference between neighboring cells, and the window specifies the size of the structuring element B used in the morphological opening. The threshold establishes the maximum allowable height difference between the ground surface and ground points. Finally, scalar adaptively modifies the threshold based on local terrain variability¹⁵.

In this study, we applied the SMRF using the SMRF library (ver. 1.1.0) in Python. By referring to Pingel et al.¹⁵, we fixed Cell as 1 m to reduce computing time, and we tested window, slope, scalar, and threshold, as shown in Table 2.

Accuracy assessment

Two different accuracy assessment methods were applied to the water and ground filtering. Water area filtering was evaluated by determining whether each point was True or False. The accuracy of the NDWI-based water filtering was assessed using type 1, type 2, and total errors, as defined in equations (7), (8), and (9). These indices were calculated for each measured cross-section using the water boundary data.

$$\text{Type 1 error} = \frac{FP}{FP + TN} \quad (7)$$

$$\text{Type 2 error} = \frac{FN}{FN + TP} \quad (8)$$

$$\text{Type error} = \frac{FP + FN}{TP + TN + FP + FN} \quad (9)$$

where TP is a True Positive that is well classified as a non-water point, TN is a True Negative that is well classified as a water point, FP is a False Positive that is misclassified as a non-water point, and FN is a False Negative that is misclassified as a water point.

The ground filtering accuracy was assessed using the MAE and RMSE to evaluate the vertical accuracy, as defined in equations (10) and (11). The measured elevation value Z_i^{measured} was obtained by linearly interpolating the 1-D elevation values measured during the cross-sectional survey using EDM. The data were divided into 1,024 points to obtain the elevation values. The calculated value $Z_i^{\text{calculated}}$ represents the elevation value from the TIN-interpolated DTM, calculated based on the previously obtained points. MAE and RMSE were analyzed across the total, vegetation, ground, and water areas, where boundary points were recorded during the EDM survey. The method with the lowest MAE and RMSE was identified as the most accurate.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i^{\text{calculated}} - y_i^{\text{measured}}| \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \left(\sum_{i=1}^n (y_i^{\text{calculated}} - y_i^{\text{measured}})^2 \right)} \quad (11)$$

where n means the number of point clouds.

Result and discussion

NDWI water filtering

Fig. 3 shows the NDWI values of all pixels calculated using Pix4D. Most of the ground and vegetation areas exhibited negative values, represented in blue, whereas waterway areas exhibited positive values, represented in red. Using the threshold value of 0.19 proposed by Xu²⁷, water areas were successfully filtered, as shown in Figure 4. Meanwhile, certain areas, such as agricultural fields or buildings located outside the stream boundary, exhibited NDWI values that exceeded 0.19. These areas were manually excluded as they were outside the scope of this study. The Type 1, Type 2, and total errors for water areas were 1.2%, 3.0%, and 2.7%, respectively, which were similar to those reported by Xu²⁷, with values of 4.14%, 0.13%, and 3.29%, respectively.

Ground filters application and accuracy analysis

MAE and RMSE with cross-section analysis were used to evaluate the accuracy of the LiDAR and SfM point cloud integrated DTM. Additionally, DTMs generated without NDWI water filtering by applying ground filters for LiDAR and SfM separately were assessed in order to confirm the effectiveness of the integration method.

Parameter optimization for each filter

Table 3 presents the optimized parameter values for each filter applied to the LiDAR and SfM point clouds. For generating DTMs using only LiDAR and SfM, LiDAR applications resulted in lower parameter values for the Resolution of CSF, Step and Spike in PTIN, and Window in SMRF compared to SfM. This is because LiDAR can generate point clouds beneath vegetation canopies by capturing the last return, which includes fewer non-ground objects and requires lower parameter values.

When generating DTMs by integrating LiDAR and SfM, LiDAR required a smaller resolution in CSF and smaller step sizes in PTIN compared to SfM. These configurations increased the sinuosity of the reference surface and effectively captured complex riparian terrain. For the SMRF, where the cell size was fixed, high scalar and threshold values were selected to account for the intricate variations in ground surfaces. Conversely, for SfM point clouds, a higher resolution in CSF, larger step sizes in PTIN, and lower scalar and threshold values in SMRF were selected. These adjustments were necessary to eliminate error points in the water areas, while preserving the limited point clouds representing riverbed features.

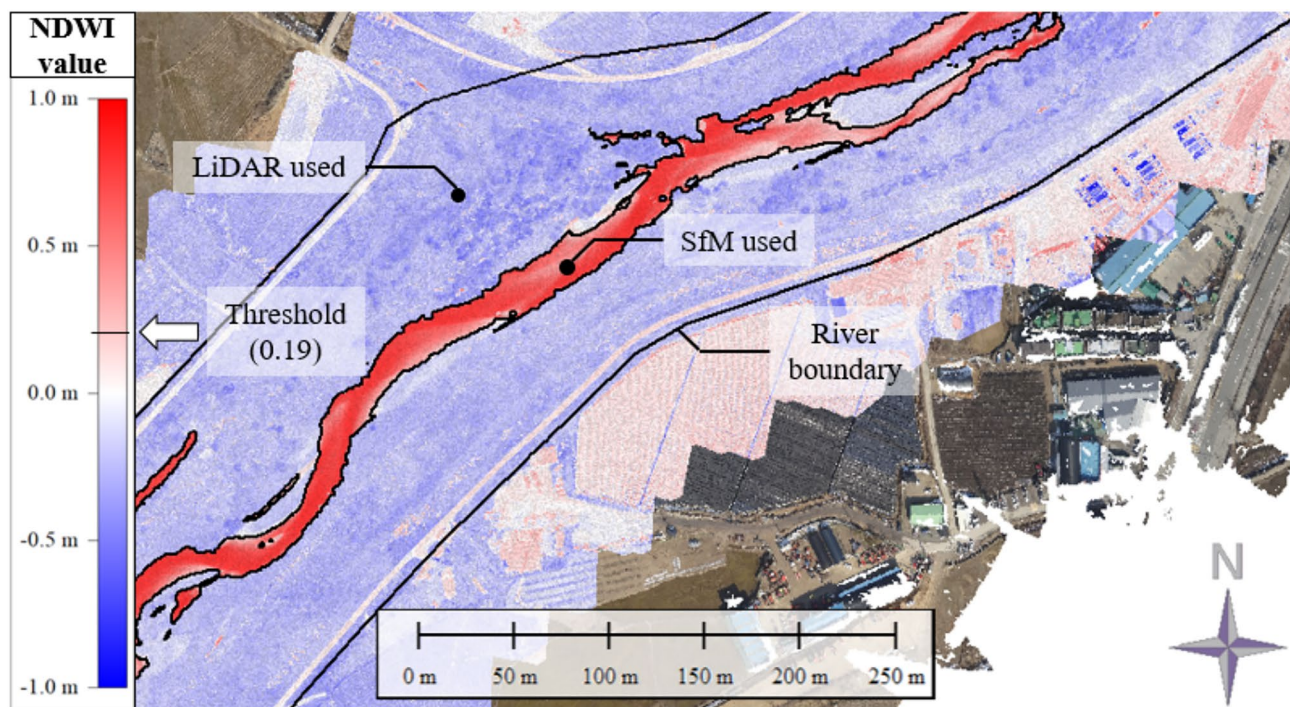


Fig. 4. Example of applying NDWI on the study site. Threshold 0.19 indicates the NDWI threshold value that differentiate water and non-water areas as introduced in Xu²⁷. The background image is drone captured orthomosaics using Phantom 4 multispectral. This map was edited using Global Mapper (Blue Marble Geographics, <https://www.blumaplegeo.com/>).

Algorithm	Parameter	LiDAR	SfM	LiDAR+ SfM	
				LiDAR	SfM
CSF	Threshold (m)	0.02	0.02	0.02	0.02
	Resolution (m)	0.2	0.4	0.2	0.9
	Rigidness (m)	1		1	
	Slope processing	FALSE		FALSE	
	Max iteration	500 times		500 times	
PTIN	Step (m)	2	3	2	5
	Spike (m)	0.2	0.4	0.2	0.2
	Offset (m)	0.02	0.02	0.08	0.02
	Bulge (m)	1		1	
	Stddev	FALSE		FALSE	
	Sub	extra fine		extra fine	
SMRF	Cell (m)	1		1	
	Window (m)	1	5	1	1
	Slope	0.02	0.02	0.02	0.02
	Scalar	0.5	0.3	0.5	0.2
	Threshold (m)	0.01	0.02	0.05	0.01

Table 3. Optimal parameter values of ground filtering algorithms. Parameters were optimized for three cases, LiDAR, SfM, and LiDAR-SfM integration. LiDAR + SfM indicates generating DTM using non-water areas in LiDAR, and water areas in SfM point clouds.

The parameter sets for LiDAR, when optimized for the entire area and for non-water areas, were similar, except for the offset in PTIN and threshold in SMRF. Because water areas occupied only 15.22% of the total area, their influence on most parameters was limited. However, the offset and threshold values decreased when water areas were included, as using only LiDAR and SfM, because smaller thresholds and offsets were required to remove as many points on the water surfaces as possible.

In contrast, the parameter sets for SfM differed significantly between optimizing the entire area and the water area. For CSF and PTIN, the resolution and step values, which determine the distance between nodes of the reference surface (cloth in CSF and TIN in PTIN), were higher for water areas than for entire areas. Because the cell size of the SMRF was fixed for computational efficiency, other parameters related to morphological operations were reduced. These differences stem from the irregular error points in water areas, where parameters that eliminate more points are necessary to enhance the accuracy.

Accuracy analysis for using only LiDAR and SfM point clouds each

Table 4 lists the MAE and RMSE values for each filter. The SfM-based DTM was more accurate than the LiDAR based on total MAE and RMSE. The most accurate DTM was generated by applying SMRF to the SfM point cloud, achieving a MAE of 0.173 m and an RMSE of 0.222 m. The main reason for the difference in accuracy between LiDAR and SfM lies in the water areas. Because LiDAR pulses cannot penetrate water surfaces⁸, LiDAR was unable to accurately generate DTM for riverbeds. In contrast, SfM partially generates point clouds beneath the water surface. Although LiDAR exhibited higher accuracy than SfM in vegetated areas, this advantage was not sufficient to overcome its inferior performance in water areas.

Accuracy analysis for integration of LiDAR and SfM point clouds

Figure 5 presents the results of CSF, PTIN, and SMRF for Sections 3, 7, 9, and 11, demonstrating the distinct characteristics of each filtering method across various terrain types, including highly vegetated areas, bare ground, and water surfaces. Among these, the SMRF exhibited the best overall performance, particularly in non-water areas.

In non-water areas, the SMRF produced results that were more consistent with the measured cross-sections compared to the other methods. In sections 7 and 11, where low vegetation is prevalent, the cross-sections generated by SMRF were closer to the measured cross-sections than those generated by the other methods. By applying SMRF to LiDAR point clouds, this method effectively addressed the common limitation of omission errors in low shrubs, as highlighted in the black box in Figure 5 (b)⁴⁰. Additionally, SMRF successfully preserved steep and terraced slopes by utilizing optimized parameters, which was also identified as one of the challenges often faced in filtering methods³⁷. In Section 9, which features dense vegetation on the left side (Figure 5 (c)), all three filters produced effective vegetation removal, consistent with previous studies suggesting that filtering methods are well suited for removing irregularly shaped high vegetation⁴⁰.

However, the SMRF showed limitations in areas dominated by ground surfaces and low-structured objects, as discussed in Section "Result and discussion". Notably, on the left side of Section "Result and discussion", SMRF produced DTMs that were positioned lower than the ground truth, leading to commission errors (Figure 5 (a)). This issue arises because of the difficulty in distinguishing between ground and vegetation areas with similar structural characteristics.

In water areas, DSM showed high elevation in certain regions, which is mainly due to riparian vegetation, reflection on the water surface, and floating debris. However, all algorithm effectively eliminated those non-ground points. Meanwhile SfM often generates erroneous points with elevations significantly lower than the actual riverbed due to matching failures. Since ground filters generally identify local minima as ground, these low outliers can be mistakenly classified as ground points. However, CSF was most robust for those error points, and SMRF was next.

In the black circle in Figure 5 (c), both the SMRF and CSF effectively eliminated error points in water areas and generated cross-sections that closely matched the measured ones. However, in the black circles in Figure 5 (a) and (d), SMRF and PTIN were influenced by erroneous points located below the riverbed, while CSF remained unaffected. In PTIN, such outliers can strongly distort nearby triangles because the triangulated network propagates local elevation errors through connected facets, often biasing the reconstructed surface over a wide area. SMRF, which relies on morphological opening, is moderately sensitive to these anomalies, an extremely low point can affect the erosion result within its structuring window, but the impact remains spatially limited to the kernel size. In contrast, CSF minimizes error propagation because its cloth model distributes collision forces through a damped spring network, as shown in equation (2): abnormal points may trigger local contact, yet the internal rigidity and damping rapidly attenuate their effect.

(Unit: m)									
Filters	Statistics	LiDAR				SfM			
		Total	Water	Ground	Vegetation	Total	Water	Ground	Vegetation
PTIN	MAE	0.236	0.595	0.127	0.215	0.212	0.258	0.112	0.272
	RMSE	0.325	0.640	0.166	0.263	0.292	0.320	0.161	0.323
CSF	MAE	0.219	0.554	0.113	0.199	0.205	0.265	0.105	0.260
	RMSE	0.300	0.605	0.144	0.236	0.270	0.313	0.145	0.307
SMRF	MAE	0.202	0.487	0.147	0.157	0.173	0.234	0.146	0.173
	RMSE	0.281	0.507	0.176	0.204	0.222	0.288	0.187	0.215

Table 4. The result of MAE and RMSE of ground filters which are applied on LiDAR and SfM point clouds each. The boundary of water, ground and vegetation areas were recorded at the time of leveling survey.

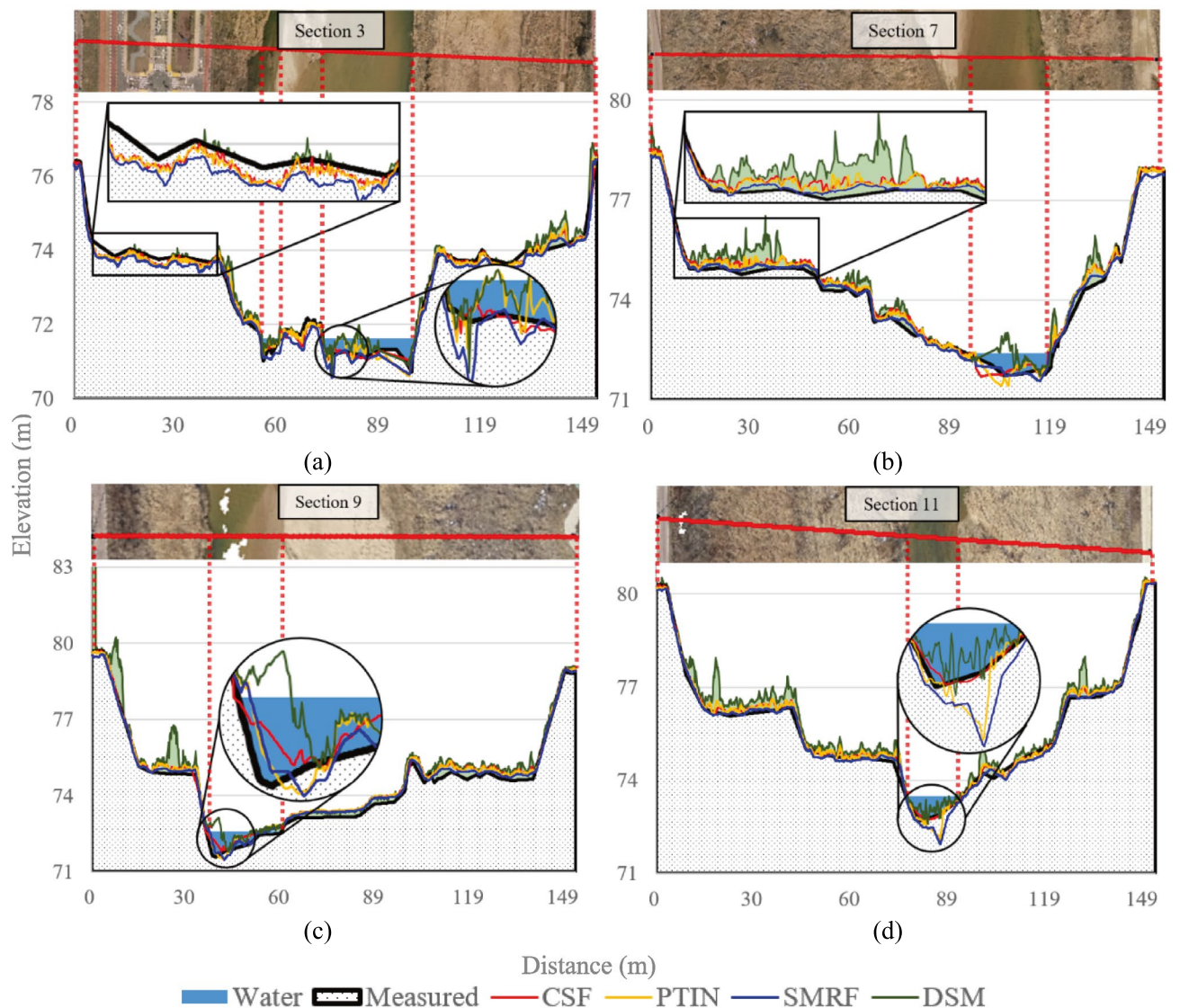


Fig. 5. Cross-section analysis of each filter, CSF, PTIN, and SMRF in section 3 (a), 7 (b), 9 (c), 11 (d). The DSM elevation, represented as green area for non-water area while highlighted as green dot line for water area, was acquired as explained in 2.2. The water elevation, expressed as light blue area, was identified during the leveling survey.

Table 5 lists the accuracy of each filter applied to the LiDAR and SfM point clouds. Among the three filters, SMRF demonstrated the best overall accuracy, particularly in vegetation areas, whereas CSF achieved the highest accuracy in ground and water areas. PTIN performed well in ground areas, but was less effective in filtering vegetation and water error points.

SMRF achieved the highest accuracy in vegetation areas, with a MAE of 0.160 m and RMSE of 0.207 m. Although SMRF showed a slightly lower accuracy than CSF in ground and water areas, it outperformed CSF in vegetation areas, leading to the lowest MAE and RMSE for the entire area (MAE = 0.160 m, RMSE = 0.214 m). The high performance of the SMRF can be attributed to its ability to retain fine terrain details, such as footpaths alongside the stream¹⁵. These findings are consistent with those of previous research, which suggests that SMRF provides more accurate results than PTIN and CSF, especially in complex terrains^{23,40}.

CSF demonstrated the highest accuracy in ground and water areas, with a MAE of 0.114 m and RMSE of 0.146 m in ground areas and an MAE of 0.213 m and RMSE of 0.253 m in water areas. However, the CSF was less effective in removing vegetation, leading to a lower overall accuracy (MAE = 0.170 m, RMSE = 0.219 m). This outcome can be attributed to CSF's robustness of the CSF in homogeneous terrains, characterized by minimal terrain fluctuations in ground and water areas. However, CSF struggle in vegetation-covered regions, including terraced terrains and shrubs^{15,16}.

PTIN also exhibited relatively high accuracy in ground areas, but failed to effectively remove vegetation and error points in water areas. Unlike SMRF, which retains detailed topographic features, PTIN's filtering approach is less efficient in distinguishing vegetation from terrain, leading to a higher rate of misclassification.

(Unit: m)					
Filters	Accuracy	LiDAR + SfM			
		Total	Ground	Vegetation	Water
CSF	MAE	0.170	0.114	0.201	0.213
	RMSE	0.219	0.146	0.239	0.253
PTIN	MAE	0.181	0.115	0.208	0.261
	RMSE	0.242	0.148	0.253	0.319
SMRF	MAE	0.160	0.135	0.160	0.225
	RMSE	0.214	0.166	0.207	0.278

Table 5. The result of MAE and RMSE which merged LiDAR and SfM point clouds, ground and vegetation area for LiDAR and water area for SfM each. The boundary of water, ground and vegetation areas were recorded at the time of leveling survey.

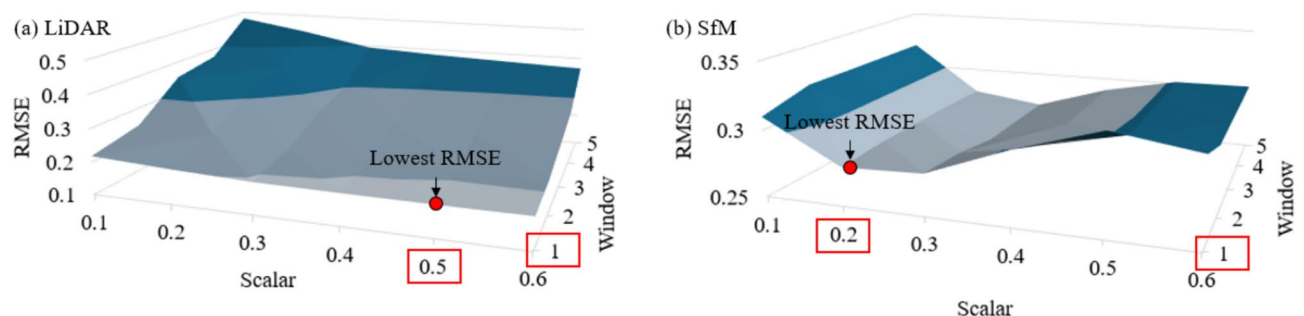


Fig. 6. Sensitivity analysis of SMRF filter parameters. The red dots highlight the optimal parameter combinations yielding the lowest RMSE.

According to table 5, SMRF filter is selected as the most suitable filter for DTM generation method using LiDAR and SfM. A sensitivity analysis was conducted to evaluate the convergence and impact of key parameters of SMRF, scalar and window, on DTM accuracy. As illustrated in Figure 6, the optimization process demonstrated that the parameters converged effectively towards optimal solutions. A distinction was observed in the magnitude of RMSE variation, in vegetation and ground areas (LiDAR), the accuracy fluctuated significantly in response to parameter changes (Figure 6 (a)), highlighting the necessity for precise tuning in complex terrain. In contrast, the water areas (SfM) exhibited relatively minor changes in RMSE across the tested range (Figure 6 (b)), indicating a lower sensitivity to parameter adjustments in smoother submerged topography. Regarding the window size, minimizing the window consistently yielded better accuracy for both datasets. However, the window size was not reduced below the grid cell resolution (1 m). When the window is smaller than the grid spacing, the morphological opening operation effectively functions as an identity operation, failing to remove non-ground features¹⁵.

Elevation discrepancy analysis

Figure 7 shows the elevation discrepancy between the DSM and DTM in the area surrounding Section 9. Elevation discrepancies were calculated by subtracting the DTM from the DSM, highlighting the variations due to misclassified vegetation and terrain features. This analysis was conducted to provide spatial differences in the filtering accuracy of the CSF, SMRF, and PTIN in different terrain types. The left side of the waterway is characterized by dense vegetation, whereas the right side consists of sandy areas and low vegetation. The left floodplain and water area were connected by a steep slope.

While all filtering algorithms effectively classified high vegetation in the upper left and upper right portions, the CSF struggled with terrain complexity on the steep slope on the left side of the waterway (Figure 7 (b)). The algorithm misclassified the edge of the slope as vegetation, leading to significant discrepancies in elevation. This issue aligns with previous findings that CSF performs well in homogeneous terrains, but struggles with abrupt elevation changes Pingel et al.¹⁵.

The SMRF demonstrated superior performance in capturing terrain details, particularly in the low vegetation area on the right side of the sandy region. Although low vegetation was visible in the orthophoto (Figure 7 (a)), both CSF and PTIN failed to filter low shrubs effectively, causing minor discrepancies. In contrast, the SMRF successfully preserved the riparian terrain while filtering the vegetation, as shown in Figure 7 (d). These results support previous research findings that SMRF excels in complex terrain filtering¹⁵. However, in the water area, SMRF exhibited greater elevation discrepancies than the other algorithms. This discrepancy suggests that SMRF may be more sensitive to outliers on water surfaces, leading to lower accuracy compared to CSF in this region.

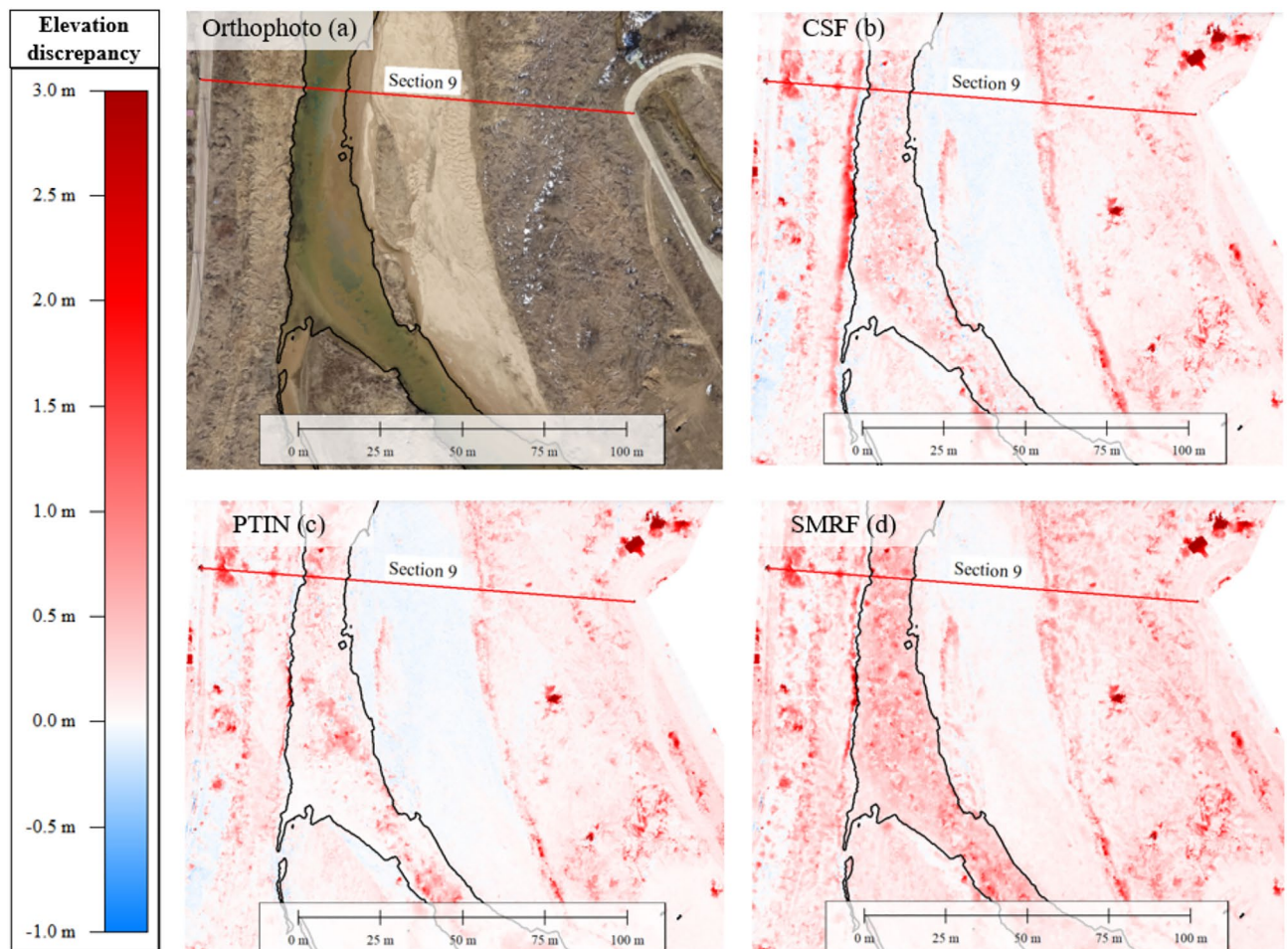


Fig. 7. Elevation discrepancy map of CSF, PTIN, and SMRF with orthophoto located around section 9. Elevation discrepancy was calculated by subtracting DSM elevation to DTM. The red area represents the positive elevation discrepancy, and the blue area represents the negative elevation discrepancy. The black line indicates the boundary of water area estimated by NDWI.

Stream DTM generation

Figure 8 presents a three-dimensional view of the final DTM generated using the SMRF filter applied to the SfM and LiDAR point clouds, along with the DSM for three selected locations (a), (b), and (c). The SMRF filter effectively removed vegetation and error points, producing a smoother and more structured DTM. The results highlight its ability to reconstruct floodplains and riverbed terrain, while also revealing limitations in water areas.

In areas A and C, which represent a vegetated floodplain and a steep slope respectively, the SMRF algorithm produced a well-structured DTM by smoothing irregular DSM features. Specifically, in area A, dense vegetation was effectively removed to create a refined terrain model, whereas in area C, DSM noise was mitigated, enhancing terrain continuity. In area B, where the DSM exhibited granular artifacts on the water surface caused by eddy currents or light reflections, the SMRF successfully removed these outliers, generating a smoother terrain representation. However, in area D, the filters failed to completely remove the bridge, retaining the bridge deck as part of the terrain. Since bridge piers are critical components for hydraulic modeling, manual post-processing is often necessary to correctly define such structures.

Conclusion

This study demonstrated that among the tested filters, SMRF was the most effective for generating DTMs from integrated UAV-SfM and LiDAR data, achieving an MAE of 0.160 m and an RMSE of 0.214 m across all areas. However, SMRF tended to easily propagate lower elevation values in water areas to adjacent points, while CSF was robust to those error points. Compared to previous approaches relying on a single data source or fixed thresholding methods, the proposed workflow, which integrates NDWI for automated classification with optimized filtering, offers improved adaptability and accuracy in complex riparian environments. The proposed method is applicable during late autumn or early spring, when vegetation vitality is low that bare ground is intermittently visible. The findings of this study can contribute to the generation of high-resolution DTMs and support mesoscale hydrological and ecological modeling in small streams. Future research should focus on

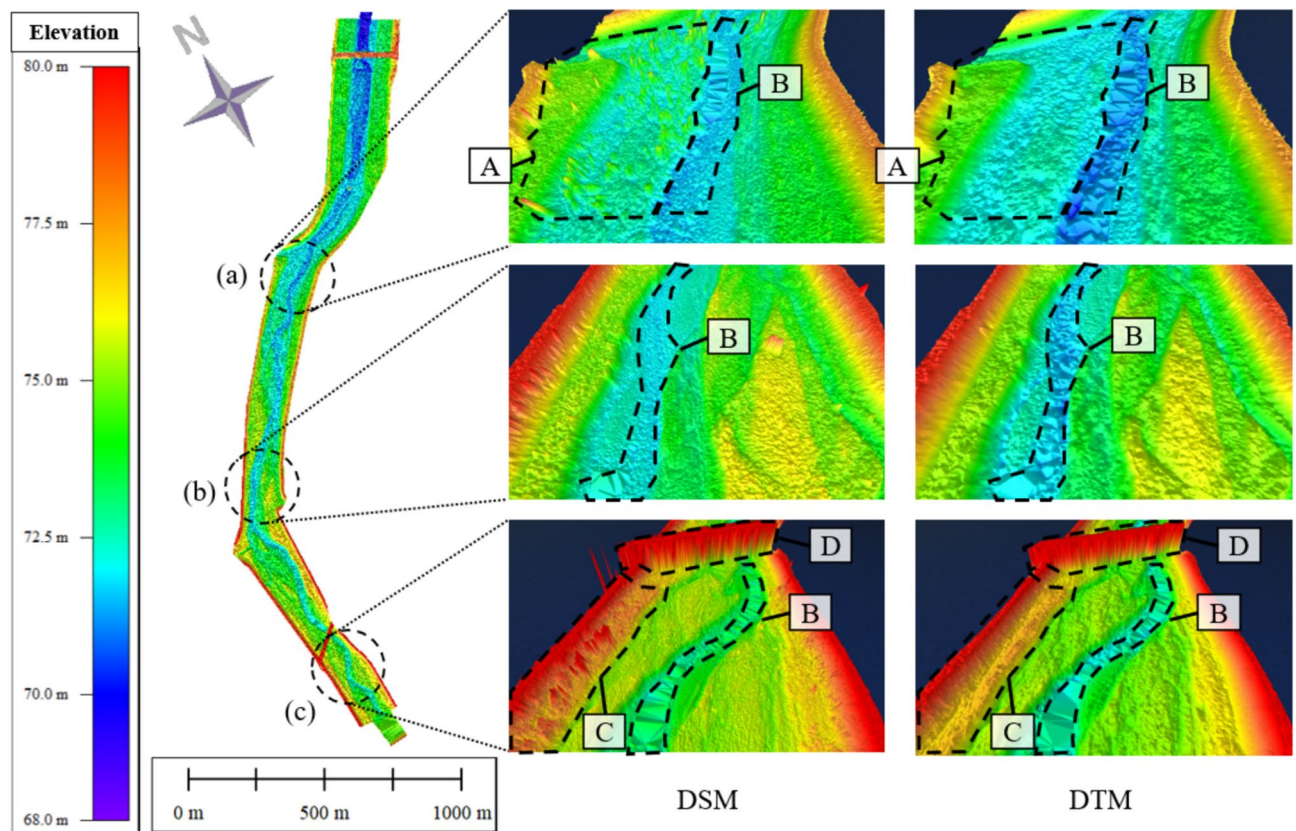


Fig. 8. Comparison of DTM and DSM for locations (a), (b) and (c). DTM was generated by integrating LiDAR and SfM using SMRF filter, and DSM is generated by selecting highest elevation point in 5 cm by 5 cm pixels. Area A represents the densely vegetated floodplain area, area B represents the water area, area C represents densely vegetated steep slope area, and area D represents the bridge.

addressing this limitation, potentially by introducing green LiDAR, and further evaluate the accuracy of the LiDAR-SfM method in shallow river areas.

Data availability

The data that support the findings of this study are available from the corresponding author, Inhong Song, upon reasonable request.

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Author contributions

Jaejun Gou conducted the measurements, performed data analysis, and drafted the manuscript. Hyeokjin Lee contributed to the measurements, assisted in developing the methodology, and supported data analysis. Jinseok Park participated in the measurements and contributed to the methodology development. Seongju Jang was involved in the measurements and supported data analysis. Nakyung Lee assisted in data analysis and contributed to the initial draft. Inhong Song provided technical insights and feedback. All authors reviewed and approved the final manuscript.

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Declarations

Conflict of interest

The authors declare no competing interests.

Additional information

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